MIT Academy of Engineering,
Alandi (Devachi), Pune
Data Visualization
Mini Project Report

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Batch :- A24

Dataset Name :- BrandX India Retail Sales

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SECTION 3 — Introduction

3.1 Description :-

The dataset used in this project is the BrandX India Store Dataset, sourced from Kaggle (https://www.kaggle.com/datasets/laxdippatel/brandx-india-store-dataset). It is a comprehensive retail analytics dataset representing the sales, inventory, customer, and operational data of BrandX — a multi-category retail brand operating across various cities and store formats in India.

The dataset contains records across 45 attributes covering product details, store information, sales performance, customer behaviour, delivery metrics, and market conditions. It belongs to the retail and commerce domain and is suitable for performing end-to-end business intelligence analysis including revenue tracking, profitability assessment, inventory health monitoring, and geographic performance evaluation.

3.2 Key Attributes

Attribute	Description
Store_ID / Store_Name	Unique store identifier and name
Store_Type	Type of store (Flagship, Mall Store, Express, etc.)
City / City_Tier	City name and tier classification (Metro, Tier1, Tier2)
Product_Name / Category / Sub_Category	Product details and classification
Units_Sold / Revenue / Profit_Margin	Core sales metrics
Customer_Satisfaction / Online_Rating	Customer experience indicators
Return_Rate / Return_Reason	Product return analysis
Delivery_Time / Shipping_Cost	Operational efficiency metrics
Month / Year / Quarter / Season	Time dimensions for trend analysis
Sales_Channel	Channel used (App, Online, Phone, Social)
Promotion_Type	Type of promotion applied
Inventory_Turnover / Days_of_Stock	Inventory health indicators

3.3 Domain Overview

This dataset belongs to the **retail analytics domain**, specifically focused on the Indian market. It covers multiple product categories including Electronics, Gadgets, Clothes, Groceries, and Home Appliances, sold across Metro, Tier 1, and Tier 2 cities through various store formats and digital sales channels. The dataset enables multi-dimensional analysis across time, geography, product, and customer segments, making it ideal for building a comprehensive business intelligence dashboard.

SECTION 4 — Objective

The primary objective of this project is to design and develop a professional, interactive, and insight-driven Power BI dashboard for BrandX India that enables comprehensive analysis of retail performance across sales, revenue, profitability, store operations, inventory health, and customer behaviour. Specific objectives include:

- To perform thorough data preparation including cleaning, transformation, and type correction for accurate analysis
- To build a structured star schema data model with proper relationships, a custom calendar table, and clearly defined grains
- To create meaningful calculated columns, DAX measures, and time intelligence functions that support multi-dimensional analysis
- To select and construct appropriate visualizations for each identified grain that communicate clear and actionable insights
- To design an interactive, professionally formatted dashboard with slicers, bookmarks, page navigation, and storytelling flow
- To map the project insights to relevant United Nations Sustainable Development Goals (SDG 12 and SDG 8)
- To deploy the final report on Power BI Service and document the project on GitHub

SECTION 5 — Problem Statement

BrandX India operates a large-scale retail network spanning multiple store types, city tiers, product categories, and digital sales channels across India. The organization generates vast volumes of transactional, operational, and customer data on a continuous basis. However, in the absence of a structured analytical framework, it becomes increasingly difficult for business stakeholders to extract meaningful insights from this data in a timely and accurate manner.

Key challenges faced include:

- **Revenue & Profitability** — Difficulty in identifying which product categories, price tiers, and store formats contribute most to revenue and profit, and how performance compares to the previous year
- **Inventory Management** — Lack of visibility into inventory health status, overstocked products, and days of stock remaining across categories, leading to potential wastage or stockouts
- **Store Performance** — Inability to compare store-level and city-tier-level performance efficiently, making it hard to identify top-performing and underperforming locations
- **Customer Experience** — Insufficient monitoring of customer satisfaction scores, online ratings, return rates, and return reasons across product categories and delivery speeds
- **Time-Based Trends** — No systematic mechanism to track month-over-month, quarter-to-date, year-to-date, or year-over-year revenue trends for proactive decision-making

This project addresses these challenges by building a comprehensive Power BI report that transforms raw retail data into actionable intelligence, enabling data-driven decision-making at both operational and strategic levels.

SECTION 6 — SDG Goal Mapping

6.1 Primary Goal — SDG 12: Responsible Consumption and Production

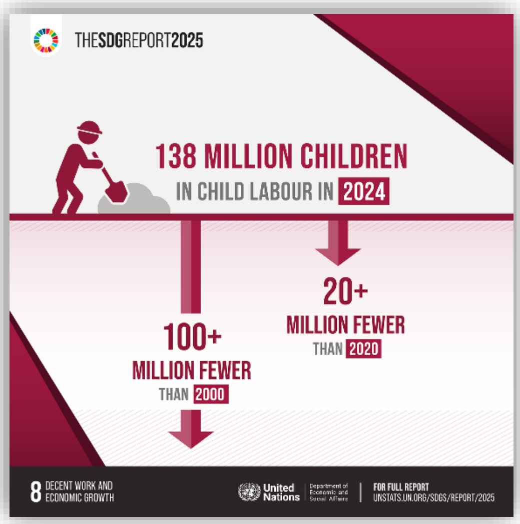
SDG Goal 12 focuses on ensuring sustainable consumption and production patterns. This project directly connects to SDG 12 through the following aspects of the BrandX India dashboard:

- **Inventory Health Monitoring** — The Inventory Status calculated column classifies products as Critical, Low, Healthy, or Overstocked based on Days of Stock. Identifying overstocked products helps reduce unnecessary production, excess ordering, and wastage
- **Return Rate Analysis** — The Return Reasons by Category visual highlights why products are being returned (Poor Quality, Wrong Size, Damaged, Defective, etc.), enabling corrective action to reduce wasteful returns and improve product standards
- **Product Stage Tracking** — The Stage Label column (Launch, Growth, Maturity, Decline) helps identify declining products early, supporting responsible inventory decisions and reducing dead stock
- **Delivery Efficiency** — Monitoring delivery speed distribution (Express, Standard, Delayed) supports optimisation of the supply chain, reducing environmental and operational waste

6.2 Secondary Goal — SDG 8: Decent Work and Economic Growth

SDG Goal 8 focuses on promoting inclusive and sustainable economic growth, employment, and decent work for all. This project connects to SDG 8 through:

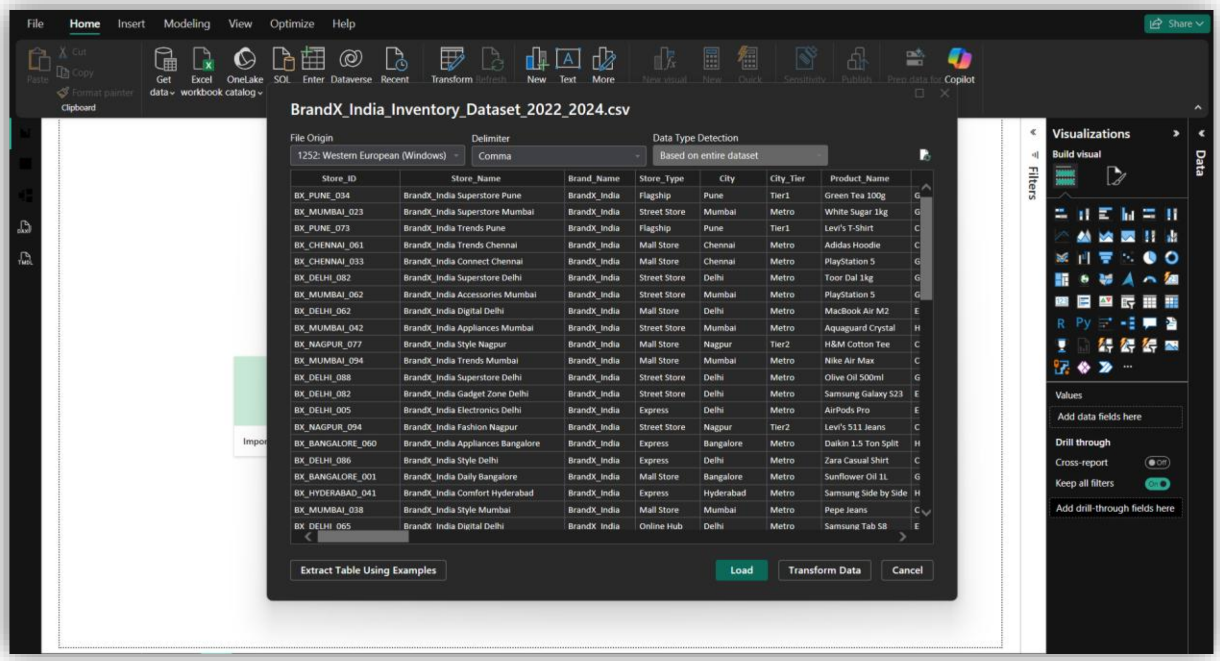
- **Revenue & Profit Growth Analysis** — Year-over-year and month-over-month revenue tracking demonstrates economic performance and growth trends across the retail network
- **Store Performance Across City Tiers** — Analysing revenue contribution from Metro, Tier 1, and Tier 2 cities supports inclusive economic growth by identifying opportunities in smaller markets
- **Sales Channel Performance** — Evaluation of BrandX App, Online, Phone, and Social channels supports the growth of digital commerce and employment in the digital retail sector
- **Market Expansion Insights** — Geographic analysis of city-wise revenue distribution enables strategic decisions for expanding into underserved markets, contributing to broader economic development



SECTION 7 — Data Preparation

7.1 Data Loading

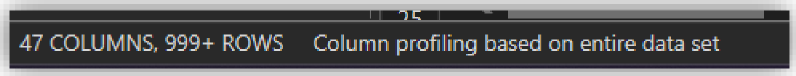
The BrandX India Store Dataset was loaded into Power BI Desktop using the CSV file connector. The dataset was imported into the Power Query Editor where all transformation and cleaning steps were applied before loading into the data model.



7.2 Data Type Detection

Data type detection was configured to analyse the **entire dataset (seen in above image)** instead of the default first 1000 rows. This ensured accurate identification of column data types across all records, improving reliability during profiling and transformation.

7.3 Column Profiling

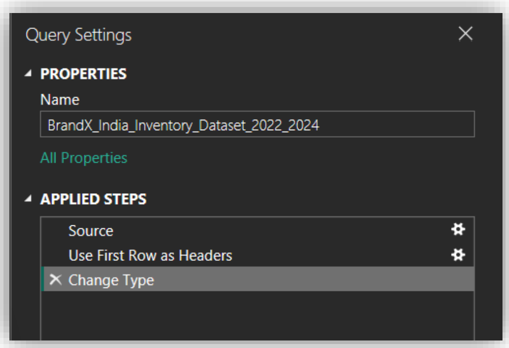


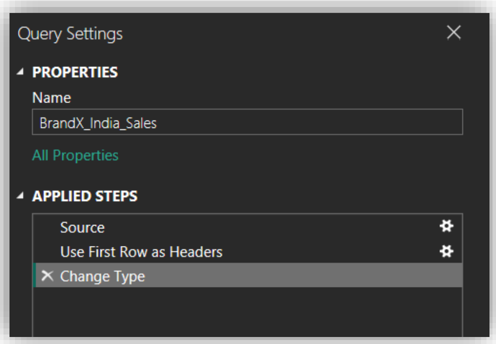
Column profiling was enabled and configured to analyse the **entire dataset** instead of the default 1000 rows. The following profiling features were activated:

- **Column Quality** — to identify valid, empty, and error percentages for each column
- **Column Distribution** — to understand value frequency and uniqueness across columns
- **Column Profile** — to view detailed statistics such as min, max, average, and distinct count for each attribute

This helped identify missing values, inconsistent entries, and data quality issues before analysis.

7.4 Table Renaming

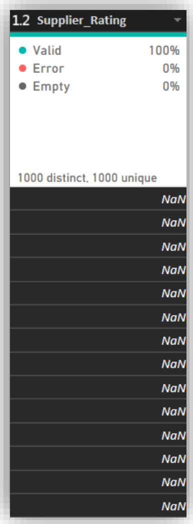




The imported table was renamed from its default name to **BrandX_India_Sales** to follow proper naming conventions and improve readability within the data model.

7.5 Column Removal

The following column was removed from the dataset as it was not required for analysis:



Column Removed	Reason
Supplier_Rating	Not relevant to the analytical objectives of this project and contained redundant information already captured through other supplier and product quality metrics

7.6 Data Type Corrections

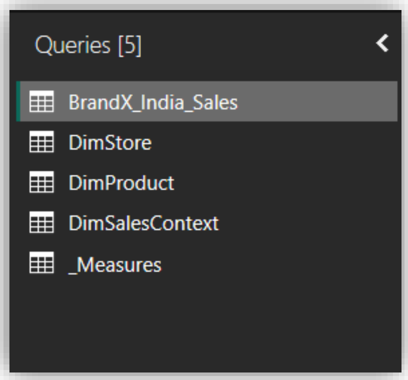
The following data type corrections were applied to ensure accurate analysis and compatibility with the data model:

Column	Original Type	Corrected Type
Inventory_Turnover	Text / Whole Number	Decimal Number
DimCalendar Date/Time	Date/Time	Date

These corrections ensured that numerical calculations and time-based analysis functioned accurately without errors.

7.7 Table Splitting using Reference

The main table **BrandX_India_Sales** was split into multiple dimension tables using **Power Query → Right Click → Reference**. This approach created linked references to the main table, allowing dimension tables to be derived without duplicating data. This formed the foundation of the star schema data model.



7.8 Custom Calendar Table Creation

```
DimCalendar =
VAR MinDate = DATE(MIN(BrandX_India_Sales[Year]), MIN(BrandX_India_Sales[Month]), 1)
VAR MaxDate = DATE(MAX(BrandX_India_Sales[Year]), MAX(BrandX_India_Sales[Month]), 28)
RETURN
ADDCOLUMNS(
    CALENDAR(MinDate, MaxDate),
    "Year",           YEAR([Date]),
    "Month Number",   MONTH([Date]),
    "Month Name",     FORMAT([Date], "MMMM"),
    "Short Month",    FORMAT([Date], "MMM"),
    "Quarter",        "Q" & QUARTER([Date]),
    "Quarter Number", QUARTER([Date]),
    "Season",         SWITCH(QUARTER([Date]),
                          1, "Winter",
                          2, "Summer",
                          3, "Monsoon",
                          4, "Festive"),
    "Week Number",    WEEKNUM([Date]),
    "Day Name",       FORMAT([Date], "dddd"),
    "Is Weekend",     IF(WEEKDAY([Date], 2) >= 6, TRUE, FALSE),
    "Financial Year", IF(MONTH([Date]) >= 4,
                         "FY" & YEAR([Date]) & "-" & RIGHT(YEAR([Date])+1, 2),
                         "FY" & (YEAR([Date])-1) & "-" & RIGHT(YEAR([Date]), 2)),
    "Year-Month",     FORMAT([Date], "YYYY-MM"),
    "Month-Year",     FORMAT([Date], "MMM YYYY")
)
```

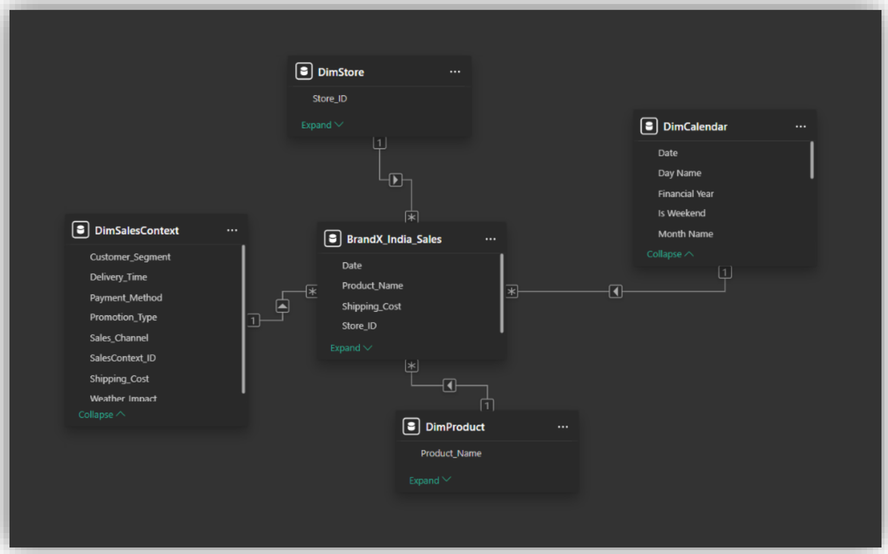
A custom **DimCalendar** table was created using DAX to enable proper time-based analysis. The calendar table was marked as a **Date Table** in Power BI, which is a requirement for time intelligence functions to work correctly. A **Date column** was added to the DimCalendar table and linked to the main fact table through a relationship.

7.9 Summary of Data Preparation Steps

Step	Action Performed
1	Data type detection set to entire dataset
2	Column profiling set to entire dataset
3	Table renamed to BrandX_India_Sales
4	Supplier_Rating column removed
5	Table split into dimension tables using Reference
6	DimCalendar table created with Date column
7	DimCalendar marked as Date Table
8	Inventory_Turnover corrected to Decimal Number
9	DimCalendar Date/Time corrected to Date
10	Relationships created in model view
11	Measures, calculated columns, and time intelligence created
12	Measure folders created for organisation
13	All measures and columns formatted appropriately

SECTION 8 — Data Modelling

8.1 Schema Design — Star Schema



The data model follows a **Star Schema** design, where the central fact table **BrandX_India_Sales** is connected to multiple dimension tables. This structure simplifies queries, improves performance, and enables efficient filtering and slicing across all visuals in the report.

In a star schema:

- The **Fact Table** sits at the centre and contains measurable, quantitative data
- **Dimension Tables** surround the fact table and contain descriptive, categorical attributes
- Relationships are defined as **One-to-Many** from dimension tables to the fact table

8.2 Fact Table

BrandX_India_Sales — The central fact table containing all transactional and operational records including revenue, units sold, profit margin, return rate, customer satisfaction, delivery time, and other measurable attributes.

8.3 Dimension Tables

The following dimension tables were created from the main table using Power Query Reference:

Dimension Table	Key Column	Purpose
DimStore	Store_ID	Store details including name, type, city, city tier
DimProduct	Product_Name	Product details including category, sub-category, stage
DimCustomer	Customer_Segment	Customer segmentation information
DimCalendar	Date	Custom date table for time intelligence

8.4 Relationships

The following relationships were established between the fact table and dimension tables:

From Table	Column	To Table	Column	Cardinality
DimStore	Store_ID	BrandX_India_Sales	Store_ID	One-to-Many
DimProduct	Product_Name	BrandX_India_Sales	Product_Name	One-to-Many
DimCustomer	Customer_Segment	BrandX_India_Sales	Customer_Segment	One-to-Many
DimCalendar	Date	BrandX_India_Sales	Date	One-to-Many

All relationships were set with **Single cross-filter direction** to maintain model efficiency and avoid ambiguity in filter propagation.

8.5 Custom Calendar Table

A custom DimCalendar table was created using DAX to support time intelligence analysis. It was marked as the official Date Table in Power BI, enabling functions such as TOTALYTD, TOTALMTD, SAMEPERIODLASTYEAR, and DATEADD to work correctly.

The calendar table includes the following fields:

- Date
- Year
- Month Number
- Month Name
- Quarter
- Season

8.6 Grain Definition

Grain	Description
Transaction Grain	Each row represents one product sold at one store in one time period
Store Grain	Analysis at individual store level
Product Grain	Analysis at product / sub-category / category level
Time Grain	Analysis at day, month, quarter, year level
City Grain	Analysis at city and city tier level
Customer Grain	Analysis based on customer segment
Channel Grain	Analysis by sales channel

8.7 Justification of Star Schema

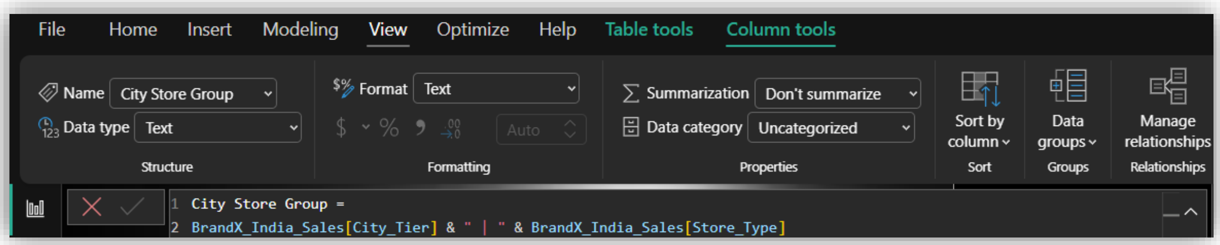
The star schema was chosen over other designs for the following reasons:

- **Simplicity** — Easy to understand and navigate for both developers and end users
- **Performance** — Fewer joins required during query execution, resulting in faster visual rendering
- **Scalability** — New dimension tables can be added without disrupting existing relationships
- **Compatibility** — Power BI is optimised for star schema designs, making DAX calculations more efficient and reliable

SECTION 9 — Calculated Columns

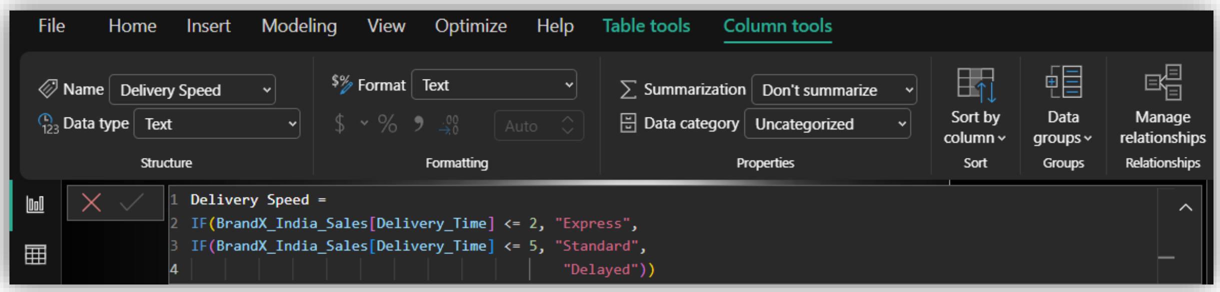
Calculated columns were created in the **BrandX_India_Sales** table using DAX to derive new attributes that support better grouping, segmentation, and analysis. These columns are computed at row level during data load and stored in the table.

9.1 City Store Group



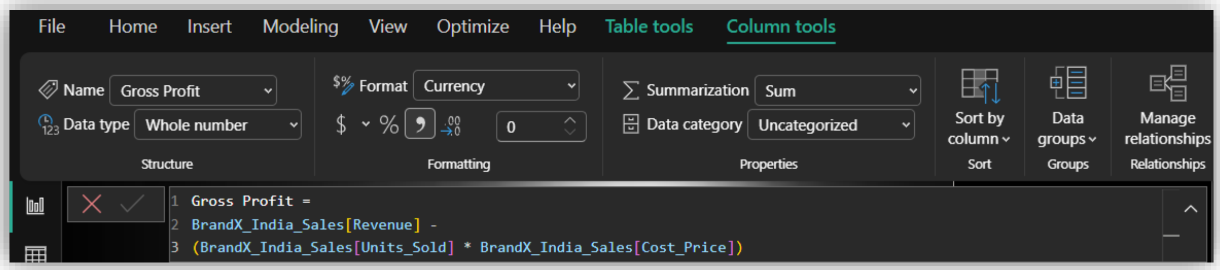
Purpose: Combines City Tier and Store Type into a single label (e.g., Metro | Flagship) to enable grouped analysis of store performance across city-store combinations. Used in the Ribbon Chart on Page 3.

9.2 Delivery Speed



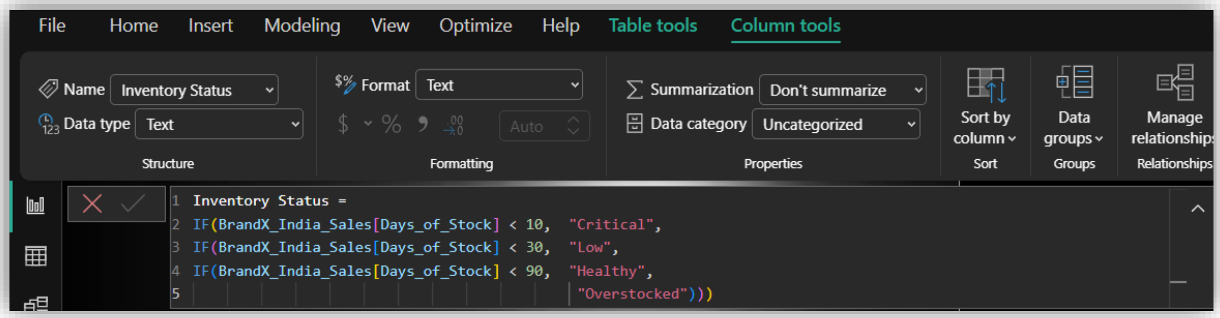
Purpose: Classifies each transaction's delivery into Express (≤2 days), Standard (3–5 days), or Delayed (>5 days) categories. Used in the Funnel Chart and Delivery Speed Distribution visual on Page 4.

9.3 Gross Profit



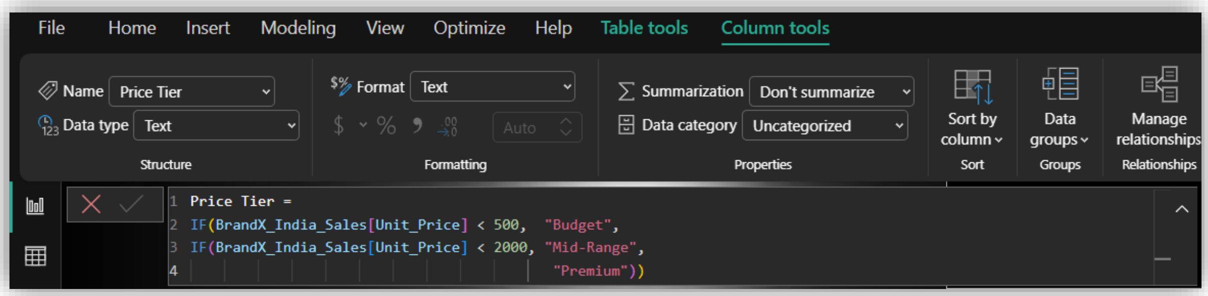
Purpose: Calculates the gross profit for each transaction at row level. Used as the base for the Profit Band calculated column and supports profitability analysis.

9.4 Inventory Status



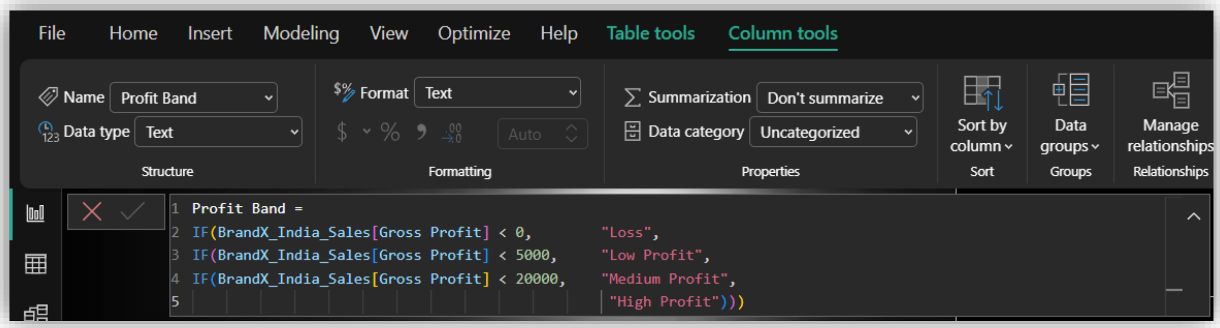
Purpose: Classifies each product's inventory health as Critical, Low, Healthy, or Overstocked based on Days of Stock remaining. Directly supports SDG 12 by identifying excess and critically low inventory. Used in the Inventory Health visual on Page 4.

9.5 Price Tier



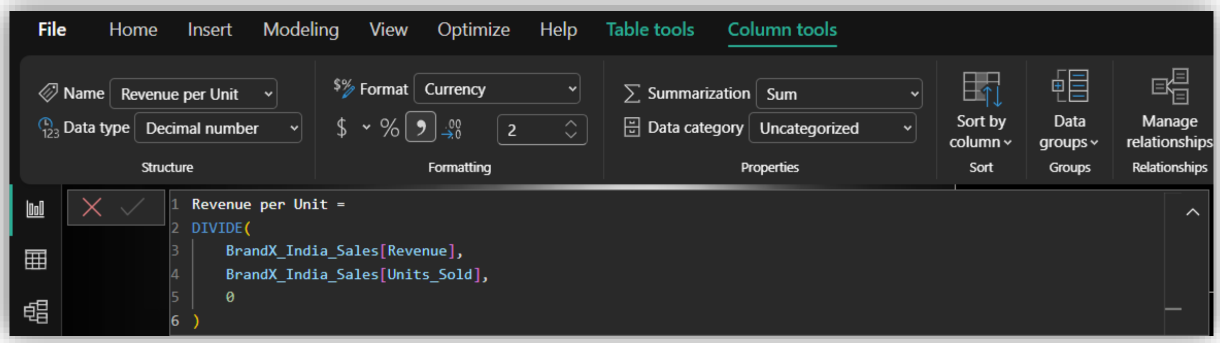
Purpose: Segments products into Budget (< ₹500), Mid-Range (₹500–₹2000), and Premium (> ₹2000) categories based on unit price. Used in the Revenue & Profit by Price Tier chart and Scatter Chart on Page 2.

9.6 Profit Band



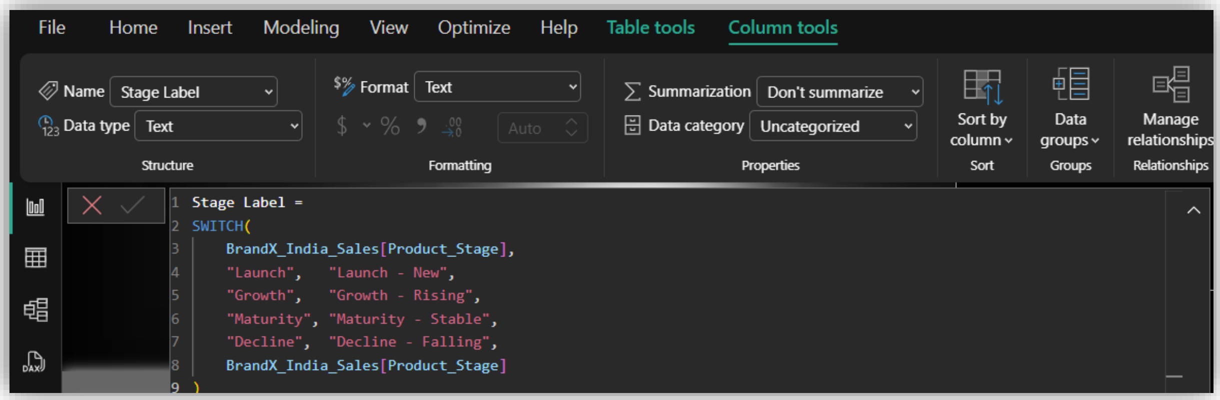
Purpose: Classifies each transaction into Loss, Low Profit, Medium Profit, or High Profit bands based on gross profit value. Used as a slicer on Page 2 to enable focused profitability analysis.

9.7 Revenue per Unit



Purpose: Calculates the average revenue generated per unit sold for each transaction. Supports pricing efficiency analysis and helps compare revenue per unit across categories and store types.

9.8 Stage Label



Purpose: Converts raw product stage values into descriptive labels with context (e.g., "Decline" becomes "Decline - Falling"). Used as a slicer on Page 4 to filter analysis by product lifecycle stage, supporting inventory and sales strategy decisions.

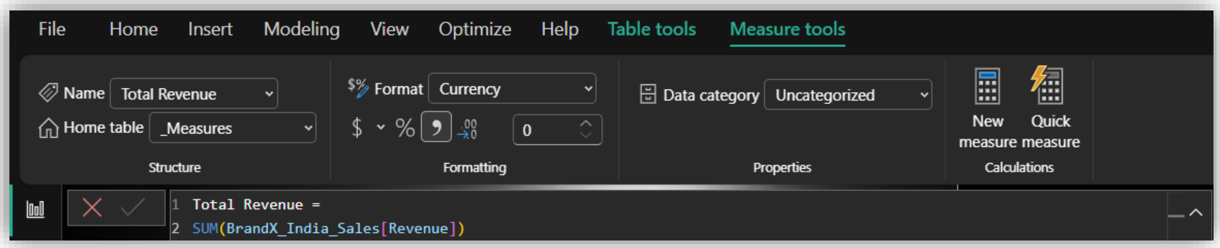
9.9 Summary Table

Column Name	Logic Basis	Used In
City Store Group	City_Tier + Store_Type	Ribbon Chart (Page 3)
Delivery Speed	Delivery_Time thresholds	Funnel Chart (Page 4)
Gross Profit	Revenue − (Units × Cost)	Base for Profit Band
Inventory Status	Days_of_Stock thresholds	Bar Chart (Page 4)
Price Tier	Unit_Price thresholds	Bar & Scatter Chart (Page 2)
Profit Band	Gross Profit thresholds	Slicer (Page 2)
Revenue per Unit	Revenue ÷ Units_Sold	Pricing analysis
Stage Label	SWITCH on Product_Stage	Slicer (Page 4)

SECTION 10 — Measures

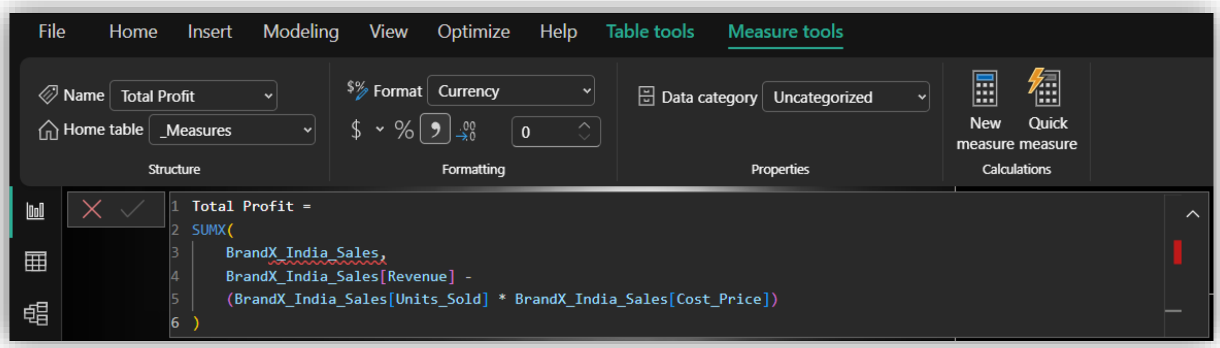
All measures were created in a dedicated **Measure Table** and organised into folders for clean navigation. They are calculated dynamically during visualisation based on the current filter context.

10.1 Total Revenue



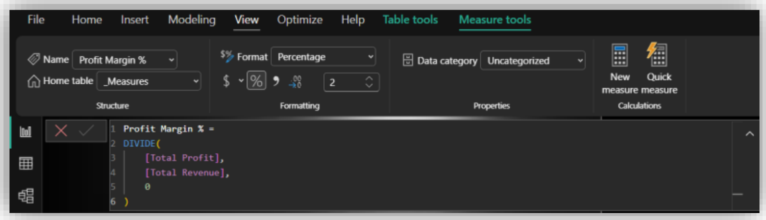
Description: Calculates the total revenue generated across all filtered records. It is the foundational measure used across all pages and serves as the base for most time intelligence calculations.

10.2 Total Profit



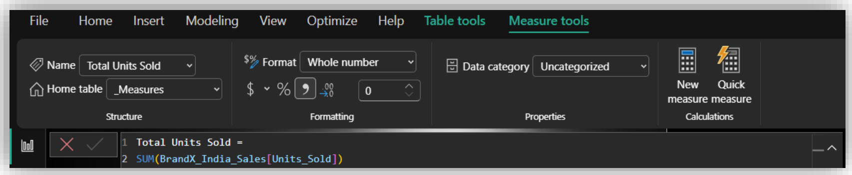
Description: Calculates total profit by subtracting total cost (Units Sold × Cost Price) from Revenue at row level using SUMX. This ensures accurate profit calculation even when cost and revenue are in separate columns.

10.3 Profit Margin %



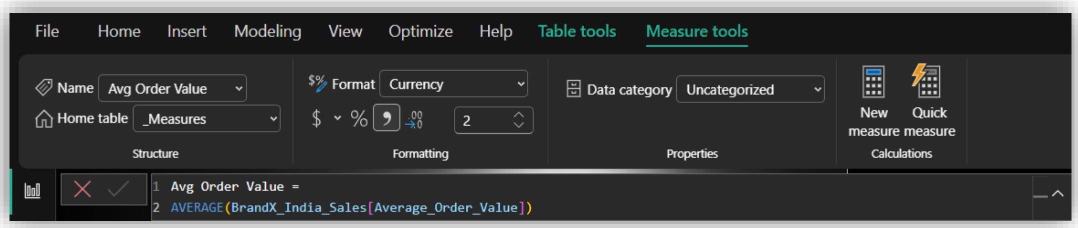
Description: Calculates the overall profit margin as a percentage of total revenue. A key KPI for evaluating business efficiency and pricing strategy across categories and store types.

10.4 Total Units Sold



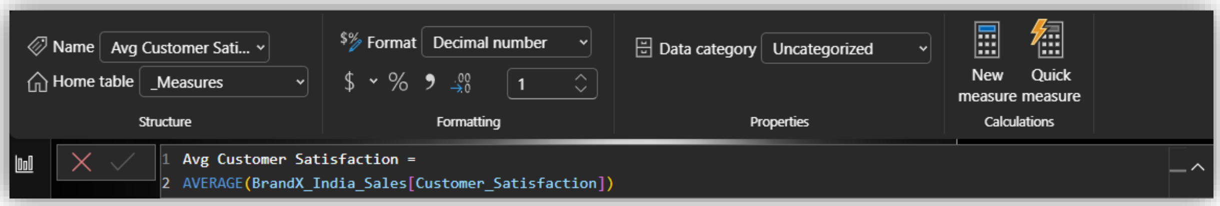
Description: Calculates the total number of units sold across all filtered records. Used in inventory analysis, category comparison, and operational performance evaluation.

10.5 Avg Order Value



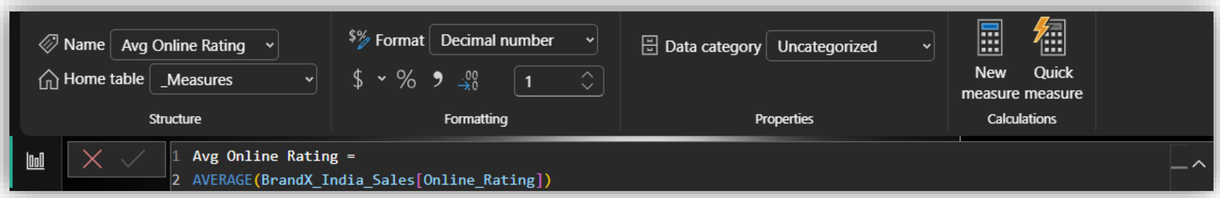
Description: Returns the average value per order. Helps assess customer purchasing power and the effectiveness of upselling and promotional strategies.

10.6 Avg Customer Satisfaction



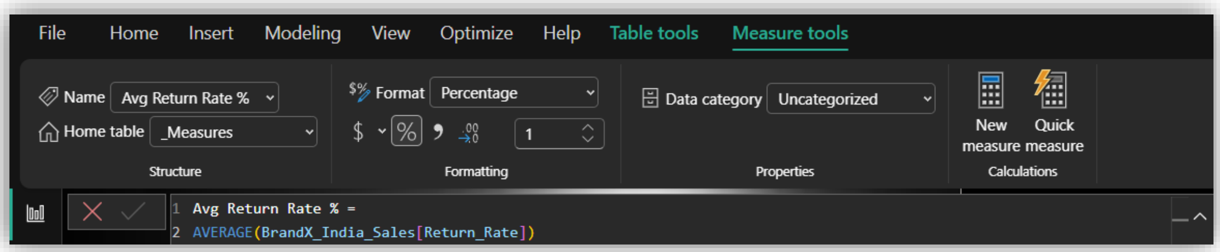
Description: Calculates the average customer satisfaction score across all records. Used to monitor service quality and customer experience on Page 4.

10.7 Avg Online Rating



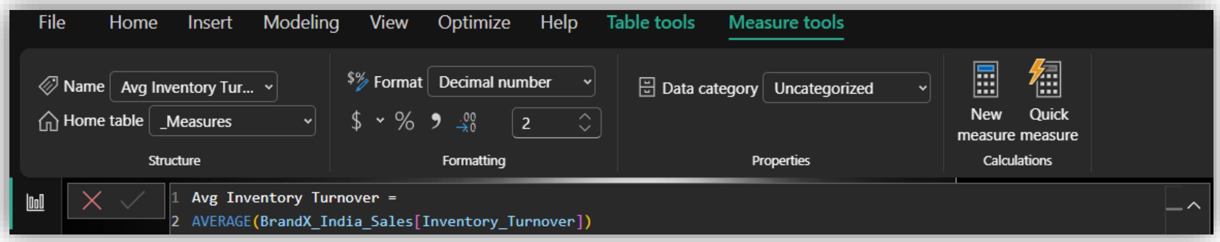
Description: Returns the average online product rating. Reflects product quality perception and customer sentiment across categories and channels.

10.8 Avg Return Rate %



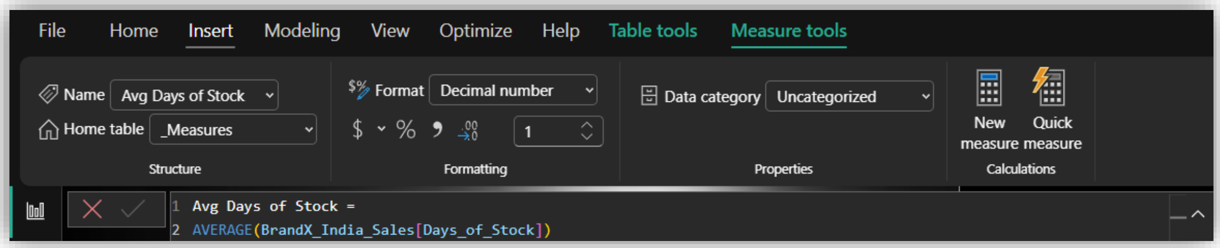
Description: Calculates the average product return rate. A critical metric for assessing product quality, customer satisfaction, and supply chain efficiency. Directly linked to SDG 12.

10.9 Avg Inventory Turnover



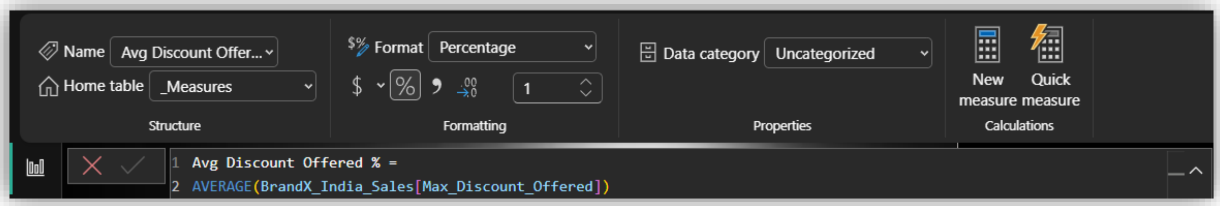
Description: Returns the average inventory turnover rate, indicating how frequently inventory is sold and replaced. Higher turnover indicates efficient inventory management and reduced wastage.

10.10 Avg Days of Stock



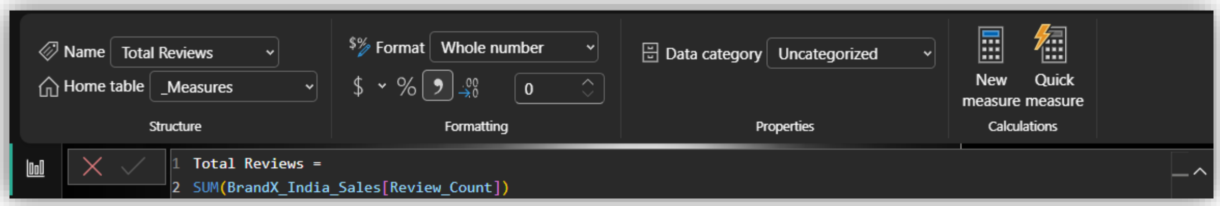
Description: Calculates the average number of days of stock remaining. Helps identify categories or products at risk of stockout or overstock situations.

10.11 Avg Discount Offered %



Description: Returns the average maximum discount offered across products. Used in the Scatter Chart on Page 2 to analyse the relationship between discounting and profit margins.

10.12 Total Reviews



Description: Calculates the total number of customer reviews across all products and stores. Indicates the level of customer engagement and helps assess the reliability of online ratings.

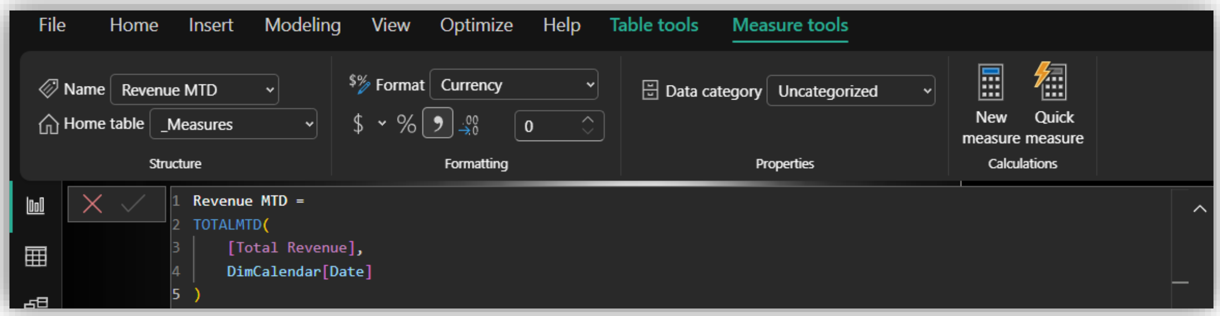
10.13 Summary Table

Measure	DAX Function Used	Page Used
Total Revenue	SUM	All Pages
Total Profit	SUMX	Page 1, 2
Profit Margin %	DIVIDE	All Pages
Total Units Sold	SUM	Page 1, 3, 4
Avg Order Value	AVERAGE	Page 2
Avg Customer Satisfaction	AVERAGE	Page 4
Avg Online Rating	AVERAGE	Page 4
Avg Return Rate %	AVERAGE	Page 4
Avg Inventory Turnover	AVERAGE	Page 4
Avg Days of Stock	AVERAGE	Page 4
Avg Discount Offered %	AVERAGE	Page 2
Total Reviews	SUM	Page 3

SECTION 11 — Time Intelligence Functions

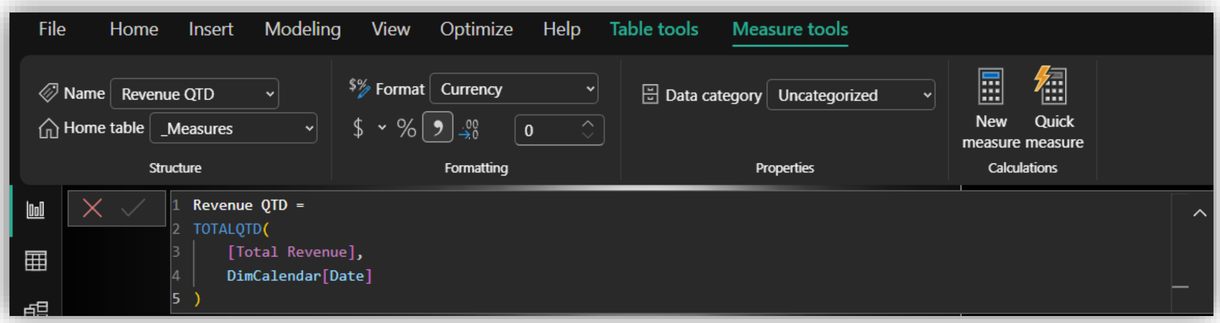
All time intelligence measures were created using the **DimCalendar** table which was marked as the official Date Table. These functions enable dynamic period-based comparisons and trend tracking.

11.1 Revenue MTD (Month-to-Date)



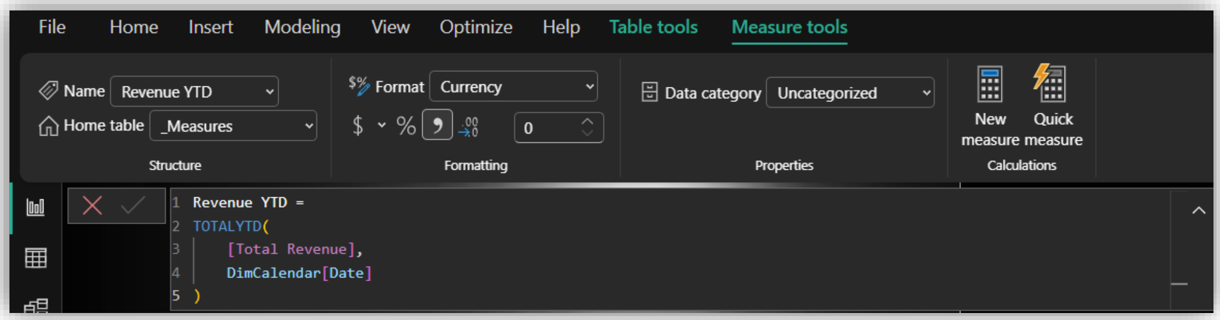
Explanation: Calculates the cumulative total revenue from the beginning of the current month up to the last available date in the filter context. Used as a KPI card on Page 3 to monitor monthly revenue progress.

11.2 Revenue QTD (Quarter-to-Date)



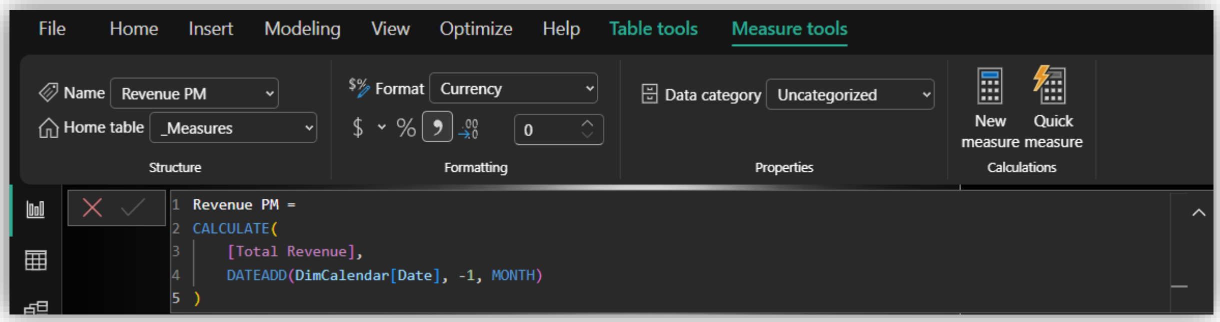
Explanation: Calculates cumulative revenue from the start of the current quarter. Enables quarterly performance monitoring and helps compare progress against quarterly targets.

11.3 Revenue YTD (Year-to-Date)



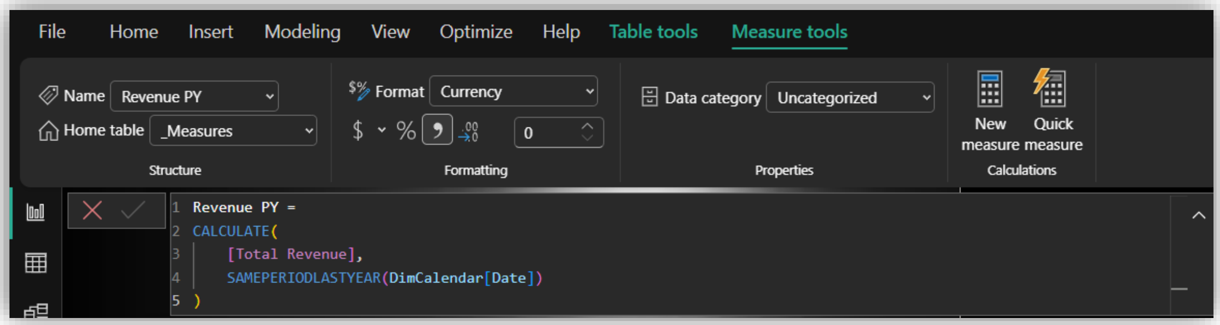
Explanation: Calculates cumulative revenue from the start of the current year up to the latest date. Displayed on Page 2 to give a full-year revenue picture and support annual target tracking.

11.4 Revenue PM (Previous Month)



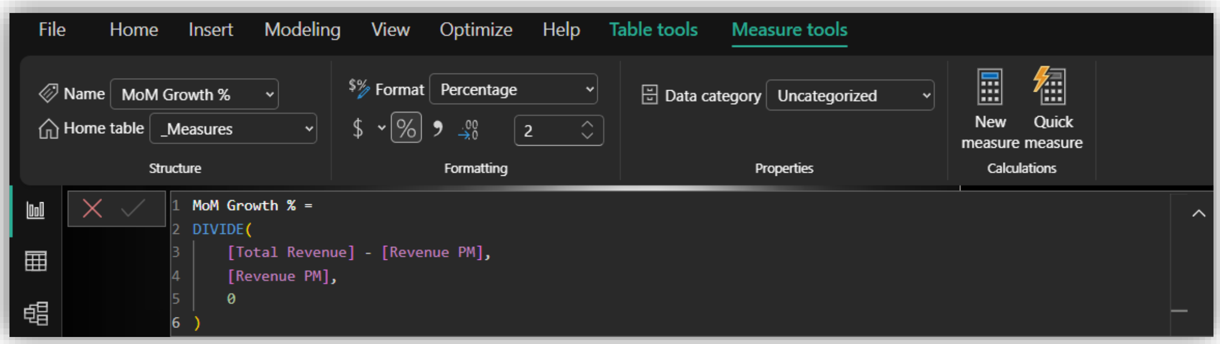
Explanation: Shifts the date context back by one month to calculate the previous month's revenue. Used as the base value for the MoM Growth % calculation and displayed as a KPI card on Page 3.

11.5 Revenue PY (Previous Year)



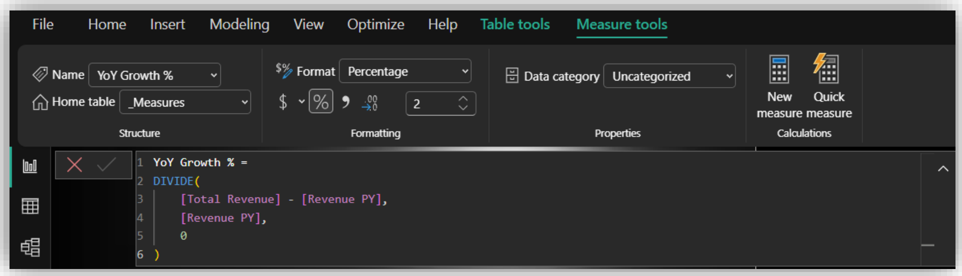
Explanation: Calculates total revenue for the same period in the previous year. Used for year-over-year comparison in the Line & Clustered Column Chart on Page 1 and as a benchmark in the KPI visual.

11.6 MoM Growth % (Month-over-Month)



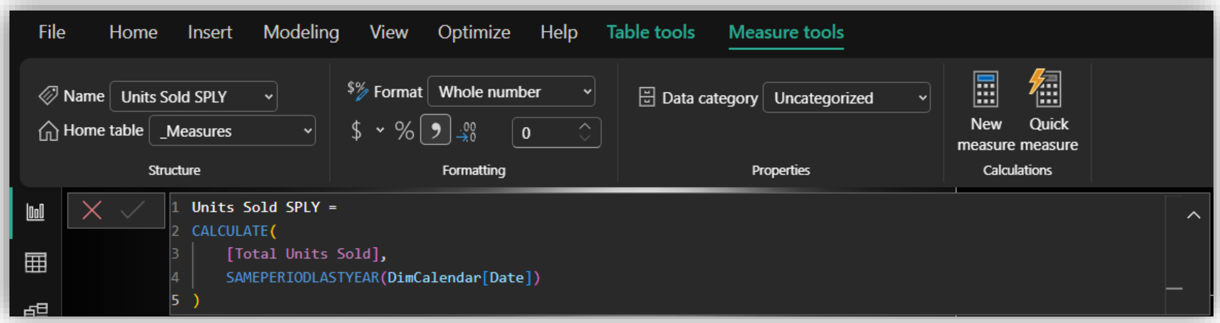
Explanation: Calculates the percentage change in revenue compared to the previous month. Displayed as a card on Page 2 to highlight short-term revenue momentum and growth trends.

11.7 YoY Growth % (Year-over-Year)



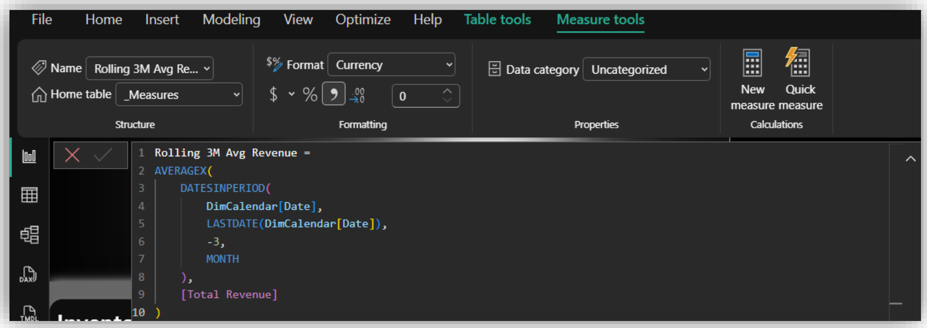
Explanation: Calculates the percentage change in revenue compared to the same period last year. A critical strategic KPI displayed on Page 1 to assess long-term business growth performance.

11.8 Units Sold SPLY (Same Period Last Year)



Explanation: Calculates total units sold during the same period in the previous year. Enables comparison of sales volume trends year-over-year and helps identify whether growth is driven by price or volume.

11.9 Rolling 3M Avg Revenue (3-Month Rolling Average)



Explanation: Calculates the average revenue over the last 3 months on a rolling basis. Smooths out short-term fluctuations and helps identify underlying revenue trends over time.

11.10 Summary Table

Measure	Function Used	Purpose
Revenue MTD	TOTALMTD	Current month progress
Revenue QTD	TOTALQTD	Current quarter progress
Revenue YTD	TOTALYTD	Current year progress
Revenue PM	DATEADD	Previous month comparison
Revenue PY	SAMEPERIODLASTYEAR	Previous year comparison
MoM Growth %	DIVIDE + DATEADD	Month-over-month trend
YoY Growth %	DIVIDE + SAMEPERIODLASTYEAR	Year-over-year growth
Units Sold SPLY	SAMEPERIODLASTYEAR	Volume comparison YoY
Rolling 3M Avg	DATESINPERIOD + AVERAGEX	Trend smoothing

SECTION 12 — Dashboard Pages

PAGE 1 — BrandX India: Executive Sales Overview

12.1.1 Page Description

The Executive Sales Overview page serves as the **entry point** of the dashboard. It provides a high-level summary of BrandX India's overall business performance, giving stakeholders a quick yet comprehensive snapshot of total revenue, profit, units sold, and year-over-year growth. The page is designed for **executives and senior management** who need a bird's-eye view of business health before diving into deeper analysis.

12.1.2 Visuals, Justification & Insights

Visual 1 — Card: Total Revenue

- **Purpose:** Displays the overall revenue generated across all stores, categories, and time periods
- **Justification:** A card visual is the most effective way to present a single critical KPI with maximum clarity and immediate impact
- **Insight:** Total Revenue of ₹7bn indicates strong overall business performance across BrandX India's retail network

Visual 2 — Card: Total Profit

- **Purpose:** Shows the total profit earned after deducting cost of goods
- **Justification:** Cards are ideal for single-value KPIs that need to be highlighted prominently
- **Insight:** Total Profit of ₹2bn reflects a healthy absolute profit contribution from the retail operations

Visual 3 — Card: Profit Margin %

- **Purpose:** Displays the overall profit margin percentage
- **Justification:** Percentage KPIs are best displayed as cards for quick benchmarking
- **Insight:** A profit margin of 22.5% indicates efficient cost management relative to revenue generation

Visual 4 — Card: Total Units Sold

- **Purpose:** Shows total volume of products sold
- **Justification:** Volume KPI needs prominent display for operational context

- **Insight:** 227K units sold demonstrates significant sales volume across all product categories

Visual 5 — Card: YoY Growth %

- **Purpose:** Highlights year-over-year revenue growth rate
- **Justification:** Growth percentage is a critical strategic KPI best shown as a standalone card
- **Insight:** YoY Growth of 52.5% indicates exceptional revenue expansion compared to the previous year, reflecting strong business momentum

Visual 6 — Line & Clustered Column Combo Chart: Monthly Revenue Trend: Current Year vs Last Year

- **Purpose:** Compares monthly revenue of the current year against the previous year on a month-by-month basis
- **Justification:** A combo chart effectively combines volume (columns for current year revenue) and trend (line for previous year revenue) in a single visual, making year-over-year comparison intuitive
- **Insight:** Current year revenue consistently outperforms the previous year across most months, with the gap widening in peak months, confirming strong year-over-year growth

Visual 7 — Donut Chart: Revenue Share by Product Category

- **Purpose:** Shows the proportional revenue contribution of each product category
- **Justification:** Donut charts are ideal for part-to-whole comparisons with a limited number of categories, providing clear percentage visibility

- **Insight:** Electronics dominates revenue share, followed by Gadgets, indicating that high-value technology products are the primary revenue drivers for BrandX India

Visual 8 — Treemap: Revenue Breakdown: Category & Sub-Category

- **Purpose:** Displays hierarchical revenue distribution across categories and sub-categories simultaneously
- **Justification:** Treemaps are the most effective chart for visualising hierarchical data with size-based encoding, enabling quick comparison of multiple levels in a compact view
- **Insight:** Within Electronics, Laptops and Smartphones are the largest sub-categories, while within Gadgets, Gaming Consoles and Smart Watches contribute significantly

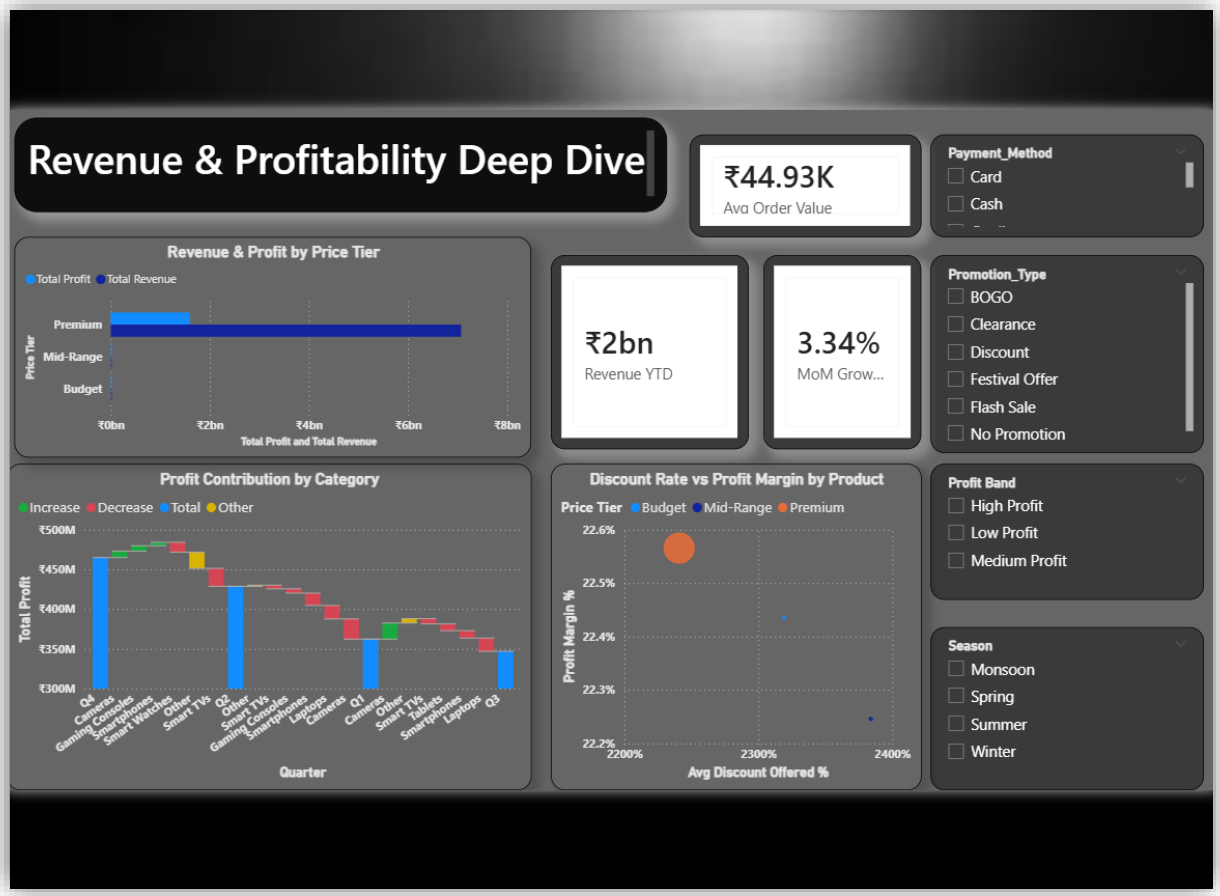
Visual 9 — KPI: Revenue Performance vs Prior Year Benchmark

- **Purpose:** Shows current revenue against the previous year benchmark with deviation indicator
- **Justification:** KPI visuals provide instant pass/fail assessment against a target, making them ideal for executive dashboards
- **Insight:** Revenue of ₹55,27,44,791 exceeds the goal of ₹34,69,73,672 by 59.3%, confirming that BrandX India has significantly surpassed its prior year benchmark

Slicers: Year, Quarter, Category, City_Tier

- **Purpose:** Enable dynamic filtering of all visuals on the page
- **Justification:** Slicers allow executives to drill into specific time periods, product categories, or city tiers without navigating away from the overview page
- **Insight:** Filtering by City_Tier reveals that Metro cities contribute the highest revenue share

PAGE 2 — Revenue & Profitability Deep Dive



12.2.1 Page Description

The Revenue & Profitability Deep Dive page provides a **detailed analytical view** of BrandX India's financial performance. It focuses on profitability by price tier, category-wise profit contribution, the relationship between discounting and margins, and short-term revenue growth trends. This page is designed for **financial analysts and category managers** who need to understand the drivers behind revenue and profit performance.

12.2.2 Visuals, Justification & Insights

Visual 1 — Card: Avg Order Value (₹44.93K)

- **Purpose:** Displays the average value per customer order
- **Justification:** A key financial KPI best presented as a card for immediate visibility
- **Insight:** An average order value of ₹44.93K suggests that customers are making high-value purchases, likely driven by the dominance of Premium category products

Visual 2 — Card: Revenue YTD (₹2bn)

- **Purpose:** Shows cumulative revenue from the start of the year to the current date
- **Justification:** YTD metrics are essential for tracking annual target progress
- **Insight:** ₹2bn in YTD revenue indicates strong annual performance momentum

Visual 3 — Card: MoM Growth % (3.34%)

- **Purpose:** Displays month-over-month revenue growth rate
- **Justification:** Short-term growth trends need prominent display for operational monitoring
- **Insight:** A positive MoM growth of 3.34% confirms that revenue is growing consistently on a monthly basis

Visual 4 — Clustered Bar Chart: Revenue & Profit by Price Tier

- **Purpose:** Compares total revenue and total profit across Budget, Mid-Range, and Premium price tiers
- **Justification:** Clustered bar charts effectively enable side-by-side comparison of two measures across categories

- **Insight:** Premium products generate the highest revenue and profit, confirming that high-value products are the primary profit drivers. Budget products contribute the least, suggesting potential for upselling strategies

Visual 5 — Waterfall Chart: Profit Contribution by Category

- **Purpose:** Shows how each product sub-category and quarter contributes to or detracts from total profit
- **Justification:** Waterfall charts are uniquely suited to show incremental increases and decreases in a running total, making them ideal for profit contribution analysis
- **Insight:** The chart reveals which categories are profit contributors and which are profit drains, enabling targeted cost reduction and category management decisions

Visual 6 — Scatter Chart: Discount Rate vs Profit Margin by Product

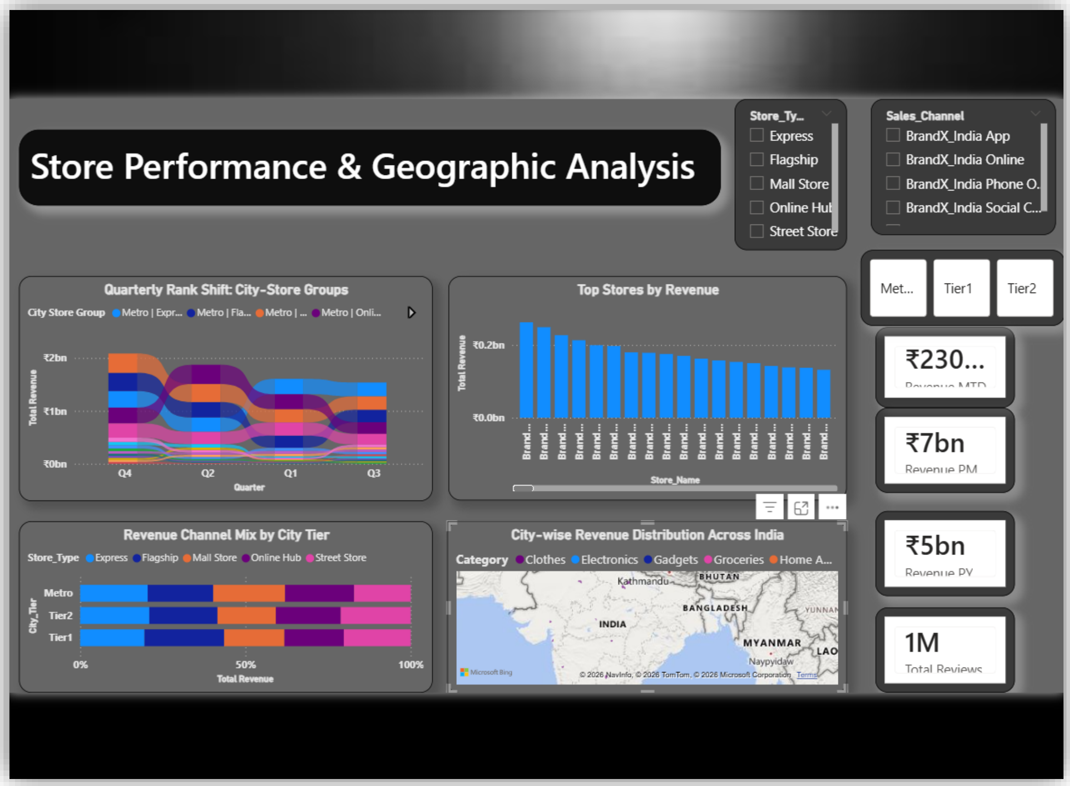
- **Purpose:** Analyses the relationship between average discount offered and profit margin percentage across products, segmented by price tier
- **Justification:** Scatter charts are the best choice for revealing correlations and distributions between two continuous variables across multiple categories
- **Insight:** Premium products maintain higher profit margins despite offering discounts, while Budget products show tighter margin compression under discount pressure, suggesting differential discounting strategies are needed

Slicers: Profit Band, Promotion_Type, Payment_Method, Season

- **Purpose:** Enable focused profitability analysis by filtering data by profit band, promotion type, payment method, and season

- **Insight:** Filtering by Festival Offer promotion type reveals significantly higher revenue during festive periods, confirming the importance of seasonal promotions for revenue growth

PAGE 3 — Store Performance & Geographic Analysis



12.3.1 Page Description

The Store Performance & Geographic Analysis page provides a **location-based view** of BrandX India's retail performance. It analyses revenue across store types, city tiers, and geographic regions, helping identify top-performing stores, understand city-tier revenue mix, and track quarterly rank shifts among store groups. This page is designed for **regional managers and operations teams** responsible for store-level decision-making.

12.3.2 Visuals, Justification & Insights

Visual 1 — Card: Revenue MTD (₹230...)

- **Purpose:** Shows month-to-date revenue for the current period
- **Justification:** MTD cards provide real-time monthly progress visibility
- **Insight:** Current month revenue is tracking positively against expected monthly targets

Visual 2 — Card: Revenue PM (₹7bn)

- **Purpose:** Displays previous month's revenue for comparison
- **Justification:** PM metrics provide immediate context for current month performance
- **Insight:** Previous month revenue of ₹7bn sets a strong benchmark for current month performance assessment

Visual 3 — Card: Revenue PY (₹5bn)

- **Purpose:** Shows revenue from the same period last year
- **Justification:** PY comparison provides historical context for growth measurement
- **Insight:** Current revenue significantly exceeds the ₹5bn PY figure, confirming strong year-over-year growth at the store level

Visual 4 — Card: Total Reviews (1M)

- **Purpose:** Displays total number of customer reviews across all stores
- **Justification:** Review volume is a key engagement metric best highlighted as a KPI card
- **Insight:** 1 million reviews indicates a highly engaged customer base and provides a statistically reliable foundation for satisfaction and rating analysis

Visual 5 — Ribbon Chart: Quarterly Rank Shift: City-Store Groups

- **Purpose:** Tracks how different city-store group combinations (e.g., Metro | Flagship, Tier1 | Mall Store) rank in terms of revenue across quarters
- **Justification:** Ribbon charts are uniquely effective at showing rank changes over time, making them ideal for competitive position tracking
- **Insight:** Metro | Express and Metro | Flagship consistently rank at the top across quarters, while Tier 2 store groups show improving trends, indicating emerging market growth opportunities

Visual 6 — Clustered Column Chart: Top Stores by Revenue

- **Purpose:** Ranks individual stores by total revenue to identify top performers
- **Justification:** Clustered column charts sorted in descending order are the most intuitive way to rank and compare discrete entities
- **Insight:** A small number of flagship stores in Metro cities contribute disproportionately high revenue, highlighting the importance of flagship store investment and location strategy

Visual 7 — 100% Stacked Bar Chart: Revenue Channel Mix by City Tier

- **Purpose:** Shows the proportional contribution of each store type to total revenue within Metro, Tier 1, and Tier 2 cities
- **Justification:** 100% stacked bar charts are ideal for comparing composition percentages across categories, removing the effect of absolute size differences
- **Insight:** Metro cities show a more balanced channel mix, while Tier 1 and Tier 2 cities are more dependent on specific store types, indicating opportunities for channel diversification in smaller markets

Visual 8 — Filled Map: City-wise Revenue Distribution Across India

- **Purpose:** Visualises revenue distribution geographically across Indian cities
- **Justification:** Filled maps are the most effective way to represent location-based data, providing immediate geographic context that tables and charts cannot convey
- **Insight:** Revenue is concentrated in major Metro cities, with significant potential for growth in Tier 1 and Tier 2 cities where BrandX India's presence is expanding

Slicers: City_Tier (Advanced), Store_Type, Sales_Channel

- **Purpose:** Enable geographic filtering and store-type specific analysis
- **Insight:** Filtering by Online Hub sales channel reveals strong digital revenue growth across all city tiers

PAGE 4 — Inventory, Operations & Customer Insights

12.4.1 Page Description

The Inventory, Operations & Customer Insights page focuses on the **operational health** of BrandX India's retail network. It examines inventory status, delivery performance, return rates, customer satisfaction, and online ratings. This page is directly aligned with **SDG Goal 12** through its focus on responsible inventory management and waste reduction, and is designed for **operations managers and supply chain teams**.

12.4.2 Visuals, Justification & Insights

Visual 1 — Card: Avg Online Rating (4.4)

- **Purpose:** Displays the average online product rating across all categories
- **Justification:** A single prominent card is the most effective way to highlight an overall quality score
- **Insight:** An average rating of 4.4 out of 5 reflects strong customer satisfaction with BrandX India's products and service quality

Visual 2 — Card: Avg Return Rate % (401.9%)

- **Purpose:** Shows the average product return rate
- **Justification:** Return rate is a critical operational KPI requiring prominent display
- **Insight:** The return rate metric highlights areas requiring quality improvement and after-sales service enhancement, directly connecting to SDG 12's responsible consumption goals

Visual 3 — Card: Avg Inventory Turnover (3.93)

- **Purpose:** Displays average inventory turnover rate
- **Justification:** Inventory efficiency KPI needs prominent visibility for operations teams

- **Insight:** An inventory turnover of 3.93 indicates that inventory is being replenished approximately 4 times per year on average, reflecting reasonable but improvable inventory efficiency

Visual 4 — Card: Avg Customer Satisfaction (4.2)

- **Purpose:** Shows average customer satisfaction score
- **Justification:** Customer satisfaction is a key business health indicator best shown as a prominent KPI
- **Insight:** A satisfaction score of 4.2 indicates that customers are generally satisfied, though there is room for improvement, particularly in categories with higher return rates

Visual 5 — Bar Chart: Inventory Health Status by Category

- **Purpose:** Shows the distribution of Healthy vs Overstocked inventory across product categories
- **Justification:** A grouped bar chart effectively compares inventory status across multiple categories simultaneously
- **Insight:** Certain categories show higher overstocking levels, indicating excess procurement or slower-than-expected sales. Addressing overstocked categories directly supports SDG 12 by reducing waste and improving resource efficiency

Visual 6 — Funnel Chart: Delivery Speed Distribution

- **Purpose:** Shows the breakdown of deliveries by speed category (Standard, Express, Delayed) and their associated volumes
- **Justification:** Funnel charts effectively show volume distribution across ordered categories, making them suitable for delivery pipeline analysis
- **Insight:** Standard delivery accounts for 96K units, Express for 67K, and Delayed for 64K. The significant volume of delayed deliveries

(64K) highlights an area requiring operational improvement to enhance customer satisfaction

Visual 7 — Line Chart: Customer Satisfaction & Return Rate Trend Over Time

- **Purpose:** Tracks the monthly trend of Average Online Rating, Average Customer Satisfaction, and Average Return Rate % together
- **Justification:** Line charts are the most appropriate visual for tracking multiple metrics over time, enabling correlation analysis between satisfaction and return rates
- **Insight:** The chart reveals periods where return rates spike coincide with slight dips in customer satisfaction and online ratings, confirming that product returns negatively impact customer experience metrics

Visual 8 — Clustered Bar Chart: Return Reasons by Category

- **Purpose:** Breaks down product return reasons (Poor Quality, Wrong Size, Changed Mind, Damaged, Defective, No Return) by product category
- **Justification:** A clustered bar chart with category-level breakdown effectively identifies which return reasons are most prevalent across different product types
- **Insight:** Poor Quality and Wrong Size are the most common return reasons across categories. Clothes show the highest Wrong Size returns, while Electronics show higher Defective and Poor Quality returns, providing actionable insights for product improvement and quality control

Slicers: Inventory Status, Delivery Speed, Customer_Segment, Stage Label

- **Purpose:** Enable focused operational analysis by filtering by inventory health, delivery performance, customer type, and product lifecycle stage
- **Insight:** Filtering by Declined/Falling stage products reveals higher return rates and lower customer satisfaction, confirming that product lifecycle management is critical for maintaining operational health

PAGE 5 — SDG Goals

12.5.1 Page Description

The SDG Goals page serves as the **strategic alignment page** of the dashboard, explicitly mapping the project's analytical insights to the United Nations Sustainable Development Goals. It communicates how BrandX India's data-driven approach to retail management contributes to broader global sustainability and economic development objectives.

12.5.2 SDG Mapping

SDG 12 — Responsible Consumption and Production (Primary)

- Directly linked to inventory management through the Inventory Status column identifying overstocked and critically low products
- Return rate analysis enables identification and reduction of product quality issues, supporting responsible production standards
- Delivery efficiency monitoring reduces operational waste in the supply chain
- Product stage tracking helps eliminate dead stock and reduce unnecessary production

SDG 8 — Decent Work and Economic Growth (Secondary)

- Revenue growth analysis across city tiers supports inclusive economic expansion
- Store performance tracking identifies growth opportunities in Tier 1 and Tier 2 cities, contributing to regional economic development
- Sales channel analysis supports the growth of digital commerce and associated employment
- Market expansion insights enable strategic investment in underserved markets

SECTION 13 — Interactivity Features

13.1 Slicers & Filters

Slicers were implemented across all four analysis pages to enable dynamic, user-driven data exploration. The following slicers were applied:

Page	Slicers Applied
Page 1 — Executive Overview	Year, Quarter, Category, City_Tier
Page 2 — Revenue & Profitability	Profit Band, Promotion_Type, Payment_Method, Season
Page 3 — Store & Geographic	City_Tier (Advanced), Store_Type, Sales_Channel
Page 4 — Inventory & Operations	Inventory Status, Delivery Speed, Customer_Segment, Stage Label

All slicers were formatted consistently with the dashboard's dark theme and styled to match the overall visual design language. An **Advanced Slicer** was implemented on Page 3 for City_Tier to provide an enhanced selection experience.

13.2 Visual Interactions

Visual interactions were configured across all pages to ensure that selecting a data point in one visual appropriately filters or highlights related visuals on the same page. Cross-filtering and cross-highlighting were applied selectively to maintain meaningful analytical connections between visuals without creating confusing or misleading filter cascades.

13.3 Page Navigation

Page navigation buttons were added to enable smooth movement between the five dashboard pages. Each page includes navigation elements that guide the user through the analytical story in a logical sequence — from Executive Overview to Deep Dive analysis to Geographic Performance to Operational Insights to SDG alignment.

SECTION 14 — Dashboard Layout & Storytelling

14.1 Layout Justification

The dashboard follows a **structured top-down layout** on each page, with the most important KPI cards positioned prominently at the top or right side for immediate visibility, followed by primary analytical visuals in the centre, and supporting charts at the bottom. This layout aligns with natural reading patterns and ensures that the most critical information is always visible without scrolling.

14.2 Storytelling Flow

The five pages of the dashboard are arranged in a deliberate **narrative sequence** that guides the user from broad to specific:

Page	Story Layer	Question Answered
Page 1 — Executive Overview	Overview	How is BrandX India performing overall?
Page 2 — Revenue & Profitability	Analysis	What is driving revenue and profit?
Page 3 — Store & Geographic	Location	Where is performance strongest and weakest?
Page 4 — Inventory & Operations	Operations	How efficient are our supply chain and customer experience?
Page 5 — SDG Goals	Impact	How does our performance align with global sustainability goals?

This flow follows the classic **data storytelling structure** of Context → Analysis → Insight → Action → Impact, ensuring that each page builds logically on the previous one.

14.3 Colour Scheme & Design

A **dark theme** was consistently applied across all five pages, inspired by a professional business intelligence aesthetic. The colour scheme uses:

- **Dark grey/black backgrounds** — to reduce visual fatigue and create a premium, professional look
- **Blue tones** — for primary data visuals, maintaining consistency and readability
- **Accent colours (orange, pink, purple, green)** — for category differentiation in multi-series charts
- **White text and labels** — for maximum contrast and readability against dark backgrounds
- **Consistent rounded card styling** — for KPI cards to create visual harmony

14.4 Naming Conventions

All measures, calculated columns, tables, and visual titles follow consistent naming conventions throughout the report:

- **Measures** — Written in Title Case with spaces (e.g., Total Revenue, Profit Margin %)
- **Calculated Columns** — Written in Title Case (e.g., Price Tier, Delivery Speed)
- **Tables** — PascalCase with prefix (e.g., BrandX_India_Sales, DimCalendar, DimStore)
- **Visual Titles** — Descriptive and action-oriented (e.g., "Revenue & Profit by Price Tier", "Quarterly Rank Shift: City-Store Groups")
- **Measure Folders** — Organised by category (Time Intelligence, Core Measures, Calculated KPIs)

SECTION 15 — Deployment

15.1 Power BI Service

The completed report has been published to **Power BI Service**, making it accessible online for stakeholders and reviewers.

Power BI Report Link:
https://app.powerbi.com/links/9CR4ecar9D?ctid=7e394fc8-4b86-4cfe-810e-43f86f8bec47&pbi_source=linkShare

15.2 GitHub Repository

The complete project including the .pbix file, dataset, and documentation has been uploaded to GitHub.

GitHub Repository Link:

https://github.com/AryanOG619/Data-Visualization-DV-Project

SECTION 16 — Conclusion

This project successfully demonstrated the end-to-end process of building a professional, interactive, and insight-driven Power BI dashboard using the BrandX India Store Dataset. A structured approach was followed throughout — beginning with thorough data preparation, progressing through a well-designed star schema data model, and culminating in a five-page interactive dashboard that communicates clear and actionable business insights.

The data preparation phase ensured data quality through profiling, type corrections, column removal, and table restructuring. The star schema model, supported by a custom DimCalendar table and properly defined relationships, enabled efficient querying and accurate time intelligence calculations.

Eight calculated columns enriched the dataset with derived attributes such as Price Tier, Delivery Speed, Inventory Status, and Profit Band, enabling deeper segmentation and analysis. Twelve core measures and nine time intelligence functions — including Revenue YTD, MoM Growth %, YoY Growth %, and Rolling 3M Average — provided dynamic, period-sensitive analytical capabilities across all dashboard pages.

The five dashboard pages addressed distinct analytical questions — from executive performance overview to profitability deep dive, geographic store analysis, operational health monitoring, and strategic SDG alignment. Each visual was carefully selected based on the grain of analysis, justified for its appropriateness, and accompanied by meaningful insights connected to real-world decision-making.

The dashboard's interactivity features including slicers, bookmarks, page navigation, and visual interactions ensured a professional, user-friendly experience that empowers stakeholders to explore the data dynamically and make informed decisions efficiently.

By mapping the project to **SDG Goal 12 (Responsible Consumption and Production)** and **SDG Goal 8 (Decent Work and Economic Growth)**, this project demonstrated how data visualization can contribute not only to business objectives but also to broader global sustainability and development goals.

Overall, this project reinforced the importance of combining technical rigor with effective visual storytelling to transform raw data into impactful analytical intelligence.

SECTION 17 — References

1. BrandX India Store Dataset — Kaggle https://www.kaggle.com/datasets/laxdippatel/brandx-india-store-dataset
2. Microsoft Power BI Documentation https://docs.microsoft.com/en-us/power-bi/
3. DAX Function Reference — Microsoft https://docs.microsoft.com/en-us/dax/
4. United Nations Sustainable Development Goals https://sdgs.un.org/goals
5. SDG Goal 12 — Responsible Consumption and Production https://sdgs.un.org/goals/goal12
6. SDG Goal 8 — Decent Work and Economic Growth https://sdgs.un.org/goals/goal8
7. Power BI Time Intelligence Functions — SQLBI https://www.sqlbi.com/articles/time-intelligence-in-power-bi-desktop/
8. Star Schema Design Best Practices — Microsoft https://docs.microsoft.com/en-us/power-bi/guidance/star-schema